



Measurement of GCase Activity in Lysates From Idiopathic and Normal Patient Fibroblast Cell Lines

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Summary (or Abstract)

Mutations in the *GBA1* gene encoding the lysosomal enzyme β -glucocerebrosidase (GCase) are associated with Parkinson's disease. These mutations lead to a reduction in GCase activity and associated with higher risk for Parkinson's disease. Parkinson patients without GBA mutations have also been reported to have reduced GCase activity suggesting that enzyme function contributes to the pathogenesis of disease (Alcalay et. Al., 2015 Brain 138:2648-2658). Our study aims to measure GCase activity from lysates prepared from idiopathic PD patient cell lines without GBA mutations compared with healthy controls. GCase activity is measured using a biochemical method with a fluorogenic substrate and detection with a fluorescence spectrophotometer.

Method

Fibroblasts were seeded in 100 mm dishes and cultured in Dulbecco's Modified Eagle's Media (DMEM) supplemented with 15% fetal bovine serum (FBS), 1% penicillin/ streptomycin (P/S), and 1% L-glutamine and incubated at 37 °C, 5% CO₂ until reaching 80-90% confluency. Cells were washed 3 × with phosphate-buffered saline (PBS), removed from plates with a cell scraper and pelleted at 2500 rpm for 5 min at 4 °C. Pellets were retained on dry ice for 10 min then lysed in McIlvaine (MI) buffer (100 mM sodium citrate, 200 mM sodium phosphate dibasic, 0.25% sodium taurocholate, and 0.1% Triton X-100, pH 5.2). Lysates were isolated after spinning at 14,000 rpm for 10 min at 4 degrees in an Eppendorf table top centrifuge. GCase activity was then measured using the substrate 4-Methylumbelliferyl- β -glucopyranoside (4-MU- β -glc ; 4-MUG). Briefly, 50 μ L of undiluted lysate is added to a 96-well black plate (Opti-





plate 96-F, Perkin-Elmer). Next, 50 μ L of 2X 4-MU- β -glc dilutions made in MI buffer from a 12 mM stock is added. Substrate made fresh on day of experiment and kept on ice. To generate a 4-MUG dose response curve, working stock solutions used were: 12, 6, 3, 1.5, 0.75, 0.375, 0 mM for final substrate concentrations of 6, 3, 1.5, 0.75, 0.375, 0.187, 0 mM. The reaction is incubated 1hr at 37 °C and quenched with 70 μ L 0.4M Glycine-NaOH, pH 10.8. Fluorescent is measured at excitation 355-360 nm, emission 450-460 nm and read on a ClarioStar Plate reader. Total protein was determined in lysates using the BioRad Protein Reagent according to the manufacturer's instructions. RFU values were normalized with protein concentration and expressed as RFU/ μ g protein. Maximal activity RFU/ μ g protein at 6 mM 4-MUGS is reported for each cell line.

References

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